CODE BUSTERS KEY

Clements Invitational

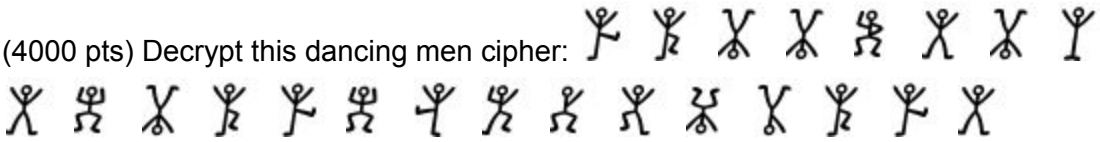
Name: _____

Team Name: _____

Team #:_____ Score: ____

1. (2000 pts) Decrypt the following Vigenere cipher using the key ANTONY: I PHAR RO ONFL AARLOE, LOG MC CPAVLS UGM.
    a. I COME TO BURY CAESAR, NOT TO PRAISE HIM.

2. (5000 pts) Decrypt the following message using mono-alphabetic substitution: PUBOTGSSQFEUZEBDUFQ
    a. DIPCHUGGETSINSPRITE

3. (4000 pts) Decrypt this dancing men cipher: 
    a. Rattle the stars if you dare

4. (4000 pts) Given the key of encryption is a= 5 and b= -2, decrypt this affine cipher: MK OQU FSYNO PQ IQGGMP PFSYKQL?
    a. IS YOU READY TO COMMIT TREASON?

5. (2000 pts) Decrypt this monoalphabetic cipher: QHE CEG WP FWZHQWEM QHIG QHE PLUMX.
    a. THE PEN IS MIGHTIER THAN THE SWORD.

6. (6000 pts) Given that the following matrix was used as the encryption matrix, decrypt: NGCH EH XBOIKBVS YRZ INGRS.

$$\begin{pmatrix} H & O \\ W & L \end{pmatrix}$$

    a. LOST IN THOUGHTS ALL ALONE.

7. (4000 pts) Given the key: ETTUBRUTE decrypt this Vigenère Cipher: RHM NIRN B PSOXX DRYLEV EXMT, SOM XLTM C MFPXH VHFY NFLX.
    a. NOT THAT I LOVED CAESAR LESS, BUT THAT I LOVED ROME MORE.

8. (4000 pts) Using the following matrix for Hill encryption, encrypt: ANY FOOL CAN CRITICIZE, CONDEMN, COMPLAIN

$$\begin{pmatrix} B & U & T \\ W & H & Y \\ T & H & O \end{pmatrix}$$

   a. ORL FYZZ WPM GFUHHNYCY, WABTWYN, JTYOHSJAV.

9. (6000 pts) Decrypt the following message using mono-alphabetic substitution: TISM PO S YSL, PD IPO RIPXD UGGQ SLQ YSNZXM GD IPO MPYX WX WJM MG OAXXV SLQ DXXQ? S WXSOM, LG YGNX.
   a. WHAT IS A MAN, IF HIS CHIEF GOOD AND MARKET OF HIS TIME BE BUT TO SLEEP AND FEED? A BEAST, NO MORE.

10. (3000 pts) Encode the following message using PINEAPPLES as the key for a Vigenère cipher: PIZZA IS A GREAT BLESSING
    a. EQMDA XH L KJTIG FLTHDMFV

11. (6000 pts) Decrypt the following Affine cipher (hint a is a number divisible by 3 less than 20): KTEQ VTFEQ QSZ KRNB, PRLZ QSZ KRNB VTFEQ.
    a. DON'T COUNT THE DAYS, MAKE THE DAYS COUNT.

12. (5000 pts) Decrypt the following Affine cipher: TG JK TGC T SFC ZTY.
    a. AN OX AND A RED CAT. Keys: a=3, b=19.

13. (3000 pts) Decrypt the following message using monoalphabetic substitution: EB NK BD IBE EB NK, EJSE HC EJK OQKCEHBI.
    a. TO BE OR NOT TO BE, THAT IS THE QUESTION.

14. (6000 pts) Decrypt the following Affine cipher: IH'A E BYDDQ PYDDQ SBTIAHKEA!
    a. IT'S A HOLLY JOLLY CHRISTMAS! Keys a=7, b=4.

Clements Invitational - Repaired practice edition - Answer key

Division C | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

8. (2 points) Authored replacement for source item Question 8: What must be known before using a Baconian table?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

A. The most common English word

B. The paper color

C. The alphabet convention

D. The carrier sentence meaning only

Correct: C